Supplementary Information

Uncertainty-aware surrogate-assisted multi-objective design of hybrid ballistic-radiological shields for RDD response

This supplement reports the complete numerical evidence summarized in the main manuscript. It is generated directly from the frozen Phase 3–5 artifacts. The 2,000 MCNP6.3 and 1,996 usable LS-DYNA training records remain part of the archived reproducibility package rather than being reproduced as manuscript tables.

S1. Complete tuned-surrogate benchmark

Every candidate was tuned by `RandomizedSearchCV` inside five-fold cross-validation on the training partition only. The table reports all 35 candidates, not only the leading models shown in the main text. PHYS-ST23 denotes the compact PHYSICS_STRENGTH23 penetration representation.

Table S1: All optimized surrogate candidates. CV quantities are means across five training folds. Inference time is the median post-fit prediction time per case on a 300-case batch.

Target	Feature set	Model	R^2	RMSE	RMSE SD	MAE	Infer. ms	Optimized hyperparameters
Dose	PHYSICS	Ridge	0.9932	0.0588	0.0017	0.0466	0.409	`alpha=0.01`
Dose	PHYSICS	Gradient Boosting	0.9908	0.0681	0.0018	0.0539	0.397	`learning_rate=0.05, max_depth=3, min_samples_leaf=2, n_estimators=700, subsample=0.85`
Dose	PHYSICS	XGBoost	0.9899	0.0714	0.0021	0.0561	0.506	`colsample_bytree=1.0, learning_rate=0.1, max_depth=3, min_child_weight=1, n_estimators=500, reg_lambda=10.0, subsample=1.0`
Dose	PHYSICS	HistGB	0.9894	0.0734	0.0032	0.0581	0.538	`l2_regularization=0.01, learning_rate=0.06, max_iter=700, max_leaf_nodes=15, min_samples_leaf=10`
Dose	PHYSICS	Extra Trees	0.9797	0.1015	0.0043	0.0814	1.257	`max_depth=24, max_features=0.7, min_samples_leaf=1, n_estimators=700`
Dose	PHYSICS	Random Forest	0.9736	0.1155	0.0062	0.0935	1.334	`max_depth=24, max_features=0.7, min_samples_leaf=1, n_estimators=700`
Dose	PHYSICS	MLP	0.9084	0.2132	0.0203	0.1683	0.470	`alpha=1e-05, hidden_layer_sizes=[128, 64, 32], learning_rate_init=0.003`
Dose	RAW	XGBoost	0.7909	0.3254	0.0174	0.2587	0.226	`colsample_bytree=0.85, learning_rate=0.1, max_depth=5, min_child_weight=10, n_estimators=500, reg_lambda=10.0, subsample=0.85`

Target	Feature set	Model	R^2	RMSE	RMSE SD	MAE	Infer. ms	Optimized hyperparameters
Dose	RAW	Gradient Boosting	0.7704	0.3411	0.0280	0.2724	0.102	learning_rate=0.08, max_depth=4, min_samples_leaf=2, n_estimators=500, subsample=0.7
Dose	RAW	MLP	0.7421	0.3570	0.0408	0.2808	0.106	alpha=0.01, hidden_layer_sizes=[128, 64], learning_rate_init=0.0003
Dose	RAW	HistGB	0.7428	0.3613	0.0221	0.2859	0.237	l2_regularization=0.01, learning_rate=0.06, max_iter=700, max_leaf_nodes=15, min_samples_leaf=10
Dose	RAW	Random Forest	0.3149	0.5896	0.0283	0.4774	1.058	max_depth=24, max_features=1.0, min_samples_leaf=1, n_estimators=700
Dose	RAW	Extra Trees	0.2826	0.6024	0.0116	0.4838	0.939	max_depth=24, max_features=0.7, min_samples_leaf=1, n_estimators=700
Dose	RAW	Ridge	0.0086	0.7091	0.0272	0.5733	0.091	alpha=100.0
Penetration	PHYS-ST23	XGBoost	0.8944	2.3153	0.2164	1.2601	0.776	colsample_bytree=0.7, learning_rate=0.04, max_depth=6, min_child_weight=10, n_estimators=300, reg_lambda=1.0, subsample=1.0
Penetration	PHYS-ST23	HistGB	0.8927	2.3328	0.2308	1.3065	0.702	l2_regularization=0.0, learning_rate=0.1, max_iter=200, max_leaf_nodes=31, min_samples_leaf=10
Penetration	PHYS-ST23	Gradient Boosting	0.8791	2.4774	0.2895	1.4588	0.585	learning_rate=0.015, max_depth=4, min_samples_leaf=4, n_estimators=700, subsample=0.85
Penetration	PHYS-ST23	Extra Trees	0.8788	2.4820	0.3479	1.2725	1.354	max_depth=24, max_features=1.0, min_samples_leaf=1, n_estimators=700

Target	Feature set	Model	R^2	RMSE	RMSE SD	MAE	Infer. ms	Optimized hyperparameters
Penetration	PHYS-ST23	Random Forest	0.8697	2.5688	0.3443	1.3377	0.894	max_depth=None, max_features=0.7, min_samples_leaf=2, n_estimators=300
Penetration	PHYSICS	Extra Trees	0.8303	2.9425	0.2951	1.5866	1.255	max_depth=24, max_features=0.7, min_samples_leaf=1, n_estimators=700
Penetration	RAW	HistGB	0.8296	2.9474	0.2698	1.8004	0.206	l2_regularization=1.0, learning_rate=0.02, max_iter=500, max_leaf_nodes=31, min_samples_leaf=10
Penetration	PHYSICS	HistGB	0.8288	2.9562	0.2015	1.7857	0.536	l2_regularization=1.0, learning_rate=0.02, max_iter=500, max_leaf_nodes=31, min_samples_leaf=10
Penetration	PHYSICS	XGBoost	0.8289	2.9603	0.2442	1.7997	0.513	colsample_bytree=0.85, learning_rate=0.04, max_depth=8, min_child_weight=10, n_estimators=500, reg_lambda=10.0, subsample=0.85
Penetration	RAW	Extra Trees	0.8176	3.0350	0.4260	1.6299	0.473	max_depth=None, max_features=0.9, min_samples_leaf=1, n_estimators=300
Penetration	RAW	Random Forest	0.8073	3.1293	0.3520	1.7326	0.894	max_depth=24, max_features=1.0, min_samples_leaf=1, n_estimators=700
Penetration	PHYSICS	Gradient Boosting	0.8039	3.1685	0.2626	2.0369	0.389	learning_rate=0.05, max_depth=4, min_samples_leaf=4, n_estimators=350, subsample=0.85

Target	Feature set	Model	R^2	RMSE	RMSE SD	MAE	Infer. ms	Optimized hyperparameters
Penetration	RAW	XGBoost	0.8034	3.1705	0.2445	2.0375	0.145	colsample_bytree=0.85, learning_rate=0.04, max_depth=8, min_child_weight=10, n_estimators=500, reg_lambda=10.0, subsample=0.85
Penetration	PHYSICS	Random Forest	0.7843	3.3258	0.3257	1.9499	1.211	max_depth=24, max_features=0.7, min_samples_leaf=1, n_estimators=700
Penetration	RAW	MLP	0.7731	3.4030	0.2887	2.3545	0.113	alpha=0.0001, hidden_layer_sizes=[64, 32], learning_rate_init=0.001
Penetration	RAW	Gradient Boosting	0.7644	3.4742	0.2694	2.2745	0.095	learning_rate=0.05, max_depth=4, min_samples_leaf=4, n_estimators=350, subsample=0.85
Penetration	PHYS-ST23	MLP	0.7126	3.8254	0.1764	2.6477	0.577	alpha=0.0001, hidden_layer_sizes=[128, 64], learning_rate_init=0.003
Penetration	PHYSICS	MLP	0.6979	3.9244	0.2715	2.7669	0.433	alpha=1e-05, hidden_layer_sizes=[128, 64, 32], learning_rate_init=0.003
Penetration	PHYS-ST23	Ridge	0.6956	3.9496	0.3613	2.8722	0.590	alpha=0.01
Penetration	PHYSICS	Ridge	0.6166	4.4329	0.3409	3.2829	0.433	alpha=0.01
Penetration	RAW	Ridge	0.5033	5.0429	0.3052	3.8291	0.090	alpha=10.0

S2. NSGA-II run-level evidence

The global union is an empirical non-dominated reference set. It is not a known physical true Pareto front. IGD and generalized spread are therefore reported for transparency, not as proof of global optimality.

Table S2: Terminal evidence for all independent NSGA-II runs.

Run	Seed	Runtime (s)	Feasible	Pareto	HV_n	HV gain	IGD	G. spread
1	42	949.7	100%	300	0.7240	0.0383	0.0289	0.5854
2	137	919.5	100%	300	0.7266	0.0409	0.0260	0.5798
3	233	1025.3	100%	300	0.7248	0.0390	0.0259	0.5457
4	349	1135.5	100%	300	0.7186	0.0329	0.0302	0.5940
5	457	1002.6	100%	300	0.7239	0.0381	0.0274	0.5022
6	563	917.1	100%	300	0.7256	0.0398	0.0258	0.5762
7	677	920.7	100%	300	0.7259	0.0402	0.0275	0.5593
8	809	986.6	100%	300	0.7236	0.0379	0.0275	0.5991
9	929	972.5	100%	300	0.7266	0.0408	0.0257	0.5909
10	1049	984.6	100%	300	0.7273	0.0416	0.0264	0.5376

S3. Complete independent high-fidelity validation

Differences are high-fidelity result minus frozen surrogate prediction. The 30 cases were selected before solver execution by objective-role and maximin-coverage rules, and are not an exchangeable random coverage sample.

Table S3: Predicted and regenerated MCNP6.3 and LS-DYNA responses for all validation candidates.

ID	Pred. dose	HF dose	Δ dose	Pred. pen.	HF pen.	Δ pen.	Upper 95% pen.	Dose PI	Pen. PI
9001	4.0728	4.2605	+0.1877	1.8261	1.4050	-0.4211	5.7596	no	yes
9002	4.3149	4.5232	+0.2082	1.7890	1.3670	-0.4220	5.7225	no	yes
9003	4.5583	4.7115	+0.1532	1.0254	1.2020	+0.1766	4.9588	no	yes
9004	4.5952	4.7458	+0.1506	1.3757	1.2990	-0.0767	5.3091	no	yes
9005	4.6428	4.7218	+0.0790	1.0973	1.1320	+0.0347	5.0308	yes	yes
9006	4.8537	5.0504	+0.1967	1.7757	1.3490	-0.4267	5.7091	no	yes
9007	4.8857	5.0059	+0.1202	4.8567	8.9090	+4.0523	8.7902	no	no
9008	4.9532	5.0959	+0.1427	1.0931	1.1190	+0.0259	5.0265	no	yes
9009	5.1289	5.1931	+0.0642	0.9822	1.0900	+0.1078	4.9157	yes	yes
9010	5.3069	5.3856	+0.0786	2.0632	2.3530	+0.2898	5.9967	yes	yes
9011	5.4118	5.5299	+0.1181	1.2746	1.1140	-0.1606	5.2080	no	yes
9012	5.4969	5.5736	+0.0766	0.5666	0.8790	+0.3124	4.5001	yes	yes
9013	5.5113	5.4532	-0.0581	1.1181	1.2020	+0.0839	5.0515	yes	yes
9014	5.7910	5.8690	+0.0780	1.1530	0.9520	-0.2010	5.0865	yes	yes
9015	5.8418	5.8562	+0.0144	1.9791	1.7320	-0.2471	5.9126	yes	yes
9016	5.8658	5.9706	+0.1048	3.9592	3.9610	+0.0018	7.8926	yes	yes
9017	5.9345	5.9558	+0.0213	7.6291	8.9060	+1.2769	11.5625	yes	yes
9018	6.1007	6.1732	+0.0725	5.6488	7.1800	+1.5312	9.5822	yes	yes
9019	6.3181	6.3320	+0.0138	1.1080	1.0660	-0.0420	5.0414	yes	yes
9020	6.3207	6.3931	+0.0725	5.7569	4.9510	-0.8059	9.6903	yes	yes
9021	6.6334	6.7280	+0.0947	18.6207	18.4850	-0.1357	22.5542	yes	yes
9022	6.6593	6.7438	+0.0845	3.9192	6.5190	+2.5998	7.8526	yes	yes
9023	6.6951	6.7073	+0.0122	7.4831	8.6930	+1.2099	11.4166	yes	yes
9024	6.8192	6.8905	+0.0713	1.2143	1.0790	-0.1353	5.1477	yes	yes
9025	7.3002	7.3786	+0.0785	19.0774	17.7200	-1.3574	23.0108	yes	yes
9026	7.4820	7.5245	+0.0425	5.3084	4.0420	-1.2664	9.2419	yes	yes
9027	7.7622	7.8751	+0.1129	6.2962	4.9520	-1.3442	10.2297	no	yes
9028	8.1524	8.2276	+0.0752	7.6811	4.7760	-2.9051	11.6146	yes	yes
9029	8.4693	8.4727	+0.0034	6.2110	6.1840	-0.0270	10.1444	yes	yes
9030	9.0093	8.9462	-0.0631	7.4862	4.8830	-2.6032	11.4197	yes	yes

S4. Targeted impact-discretization diagnostic

This post-campaign diagnostic isolates shield and FSP refinement for validation case 9012. It is reported for transparency and is not used to replace any campaign label, retrain a surrogate, or support a mesh-convergence claim. "Active-by-log" excludes a nodal peak recorded after the associated FSP node had been removed; it is a diagnostic quantity, not a validated response label.

Table S4: Case-9012 impact-discretization diagnostic. The original/original configuration is the frozen campaign model.

Shield / FSP discretization	Raw penetration (mm)	Active-by-log (mm)	Final E/E_0	Max. $\lvert E/E_0 - 1\rvert$	FSP diagnostic
Original / original	0.879	0.879	1.00537	0.00537	No recorded FSP erosion
M1 / original	3.657	3.657	1.07232	0.07375	FSP intact; shield-element failures recorded
Original / M1	128.096	0.879	0.523659	0.476357	Four of eight FSP elements failed by negative volume before the raw peak
M1 / M1	9.220	9.220	1.00087	0.00087	FSP intact

The isolations do not form a stable, monotonic refinement sequence. Shield-only refinement exceeded the pre-specified 5% energy-deviation screening gate. FSP-only refinement suffered rapid FSP erosion, making its late raw nodal peak nonphysical for the stated numerical interpretation. The coupled M1 configuration was energetically stable but materially different from the frozen campaign response. Therefore, the diagnostic identifies a model-development requirement for a future mesh study, rather than a basis for extrapolating a mesh-independent penetration value.

S4. Post-review evidence audit

These analyses were performed after coauthor review. They do not replace the frozen models or the original Phase 5 comparisons. CSV evidence and reproducible scripts accompany the revision.

Table S5: Conditional bootstrap intervals for the original test observations.

Response	Metric	Estimate	Lower 95percent	Upper 95percent
dose	r2	0.99391	0.99253	0.99503
dose	mae	0.046156	0.042285	0.050135
dose	rmse	0.057305	0.052835	0.061854
penetration	r2	0.88607	0.83528	0.9288
penetration	mae	1.1865	0.97239	1.4213
penetration	rmse	2.3065	1.759	2.8466

Table S6: Fixed-pool threshold sensitivity, without re-optimization.

Pool	Limit	N	Point feasible	Margin feasible	Removed
DoE	15	2000	1576	1281	295
DoE	20	2000	1830	1635	195
DoE	30	2000	1980	1947	33
DoE	60	2000	2000	2000	0
Pareto union	15	1530	1478	1466	12
Pareto union	20	1530	1525	1478	47
Pareto union	30	1530	1530	1530	0
Pareto union	60	1530	1530	1530	0

Table S7: Test errors and interval coverage by observed-response quartile.

Response	Observed range	N	MAE	RMSE	Coverage
dose	(4.837, 6.039]	75	0.048191	0.059756	0.94667
dose	(6.039, 6.555]	75	0.050279	0.062102	0.89333
dose	(6.555, 7.144]	75	0.043448	0.053529	0.97333
dose	(7.144, 8.549]	75	0.042706	0.053314	0.96
penetration	(0.776, 4.396]	75	0.42539	0.89702	0.98667
penetration	(4.396, 8.794]	75	0.81972	1.2331	0.97333
penetration	(8.794, 14.304]	75	1.0179	1.451	0.98667
penetration	(14.304, 36.522]	75	2.4828	4.1047	0.78667

Table S8: All ten MCNP fluctuation-chart diagnostic outcomes, 2,000 outputs. Forty-three distinct cases fail at least one check.

Check	Failed cases
Mean behavior	11
Relative error value	0
R decrease	0
R decrease rate	0
VOV value	0
VOV decrease	0
VOV decrease rate	0
FOM value	0
FOM behavior	36
PDF slope	0

S5. Complete validation designs and numerical differences

Layer sequences are written from the inner to outer surface. Material abbreviations are Al, Fe, SiC, U (UHMWPE), B (B4C-HDPE), W (W-epoxy), and Bi (Bi_2O_3-HDPE). Thicknesses follow the same order in cm. These are all thirty pre-specified candidates, not a favorable subset.

Table S9: Thirty layer configurations, solver responses, and accounting objectives.

ID	Inner-to-outer materials	Thicknesses (cm)	Dose	Pen.	Mass	Cost
9001	Bi / B / U / W / Al / SiC / Fe	0.424 / 0.214 / 0.204 / 2.000 / 0.276 / 0.883 / 1.999	4.2605	1.405	256.21	15045.1
9002	W / Bi / U / B / Al / SiC / Fe	1.543 / 0.219 / 0.220 / 0.221 / 0.212 / 1.626 / 1.959	4.5232	1.367	249.42	11286.1
9003	W / Bi / B / U / Al / SiC / Fe	1.735 / 1.420 / 0.252 / 0.201 / 0.383 / 0.355 / 1.654	4.7115	1.202	231.74	13070.4
9004	W / B / U / Bi / SiC / Al / Fe	0.969 / 0.200 / 0.200 / 1.516 / 0.927 / 0.333 / 1.855	4.7458	1.299	232.00	7519.4
9005	Bi / B / U / W / SiC / Al / Fe	1.615 / 0.216 / 0.220 / 1.946 / 0.385 / 0.206 / 1.412	4.7218	1.132	226.23	15449.3
9006	W / B / U / Bi / SiC / Al / Fe	0.344 / 0.352 / 0.339 / 1.785 / 0.735 / 0.454 / 1.991	5.0504	1.349	224.02	3150.6
9007	W / Bi / U / B / Fe / Al / SiC	0.684 / 1.976 / 0.315 / 0.258 / 1.801 / 0.548 / 0.418	5.0059	8.909	218.16	5621.3
9008	W / Bi / U / Al / B / SiC / Fe	1.537 / 1.747 / 0.275 / 0.377 / 0.357 / 0.344 / 1.363	5.0959	1.119	212.86	11720.3
9009	W / Bi / U / B / SiC / Al / Fe	1.090 / 1.952 / 0.201 / 0.201 / 0.454 / 0.818 / 1.284	5.1931	1.090	205.45	8545.6
9010	U / Bi / B / W / SiC / Al / Fe	0.306 / 1.756 / 0.221 / 1.985 / 0.723 / 0.309 / 0.700	5.3856	2.353	194.99	15893.0
9011	W / Bi / B / U / Al / SiC / Fe	0.690 / 2.000 / 0.451 / 0.397 / 0.763 / 0.341 / 1.358	5.5299	1.114	196.64	5687.8
9012	W / Bi / B / U / Fe / Al / SiC	1.497 / 1.766 / 0.292 / 0.280 / 0.786 / 0.422 / 0.957	5.5736	0.879	188.29	11428.8
9013	Al / Bi / U / B / SiC / W / Fe	1.313 / 1.908 / 0.200 / 0.385 / 0.342 / 0.200 / 1.652	5.4532	1.202	206.35	2385.0
9014	W / Bi / U / Al / B / SiC / Fe	1.176 / 1.943 / 0.282 / 0.282 / 0.983 / 0.450 / 0.884	5.8690	0.952	178.86	9172.8
9015	W / Bi / U / B / SiC / Al / Fe	0.857 / 1.991 / 0.454 / 0.441 / 1.039 / 0.375 / 0.843	5.8562	1.732	178.14	6858.8
9016	W / Bi / U / B / Al / SiC / Fe	1.706 / 1.932 / 0.293 / 0.377 / 0.380 / 0.945 / 0.367	5.9706	3.961	172.03	13013.9
9017	W / Bi / SiC / Al / Fe / B / U	0.200 / 1.807 / 0.200 / 0.230 / 1.679 / 0.200 / 1.684	5.9558	8.906	182.15	2088.2
9018	Fe / Bi / B / W / Al / SiC / U	0.209 / 1.849 / 0.209 / 1.961 / 0.807 / 0.318 / 0.647	6.1732	7.180	162.86	15742.2
9019	W / Bi / B / U / Al / Fe / SiC	0.676 / 2.000 / 0.394 / 0.371 / 0.514 / 0.458 / 1.587	6.3320	1.066	161.23	5565.8
9020	W / Bi / Al / Fe / SiC / B / U	0.200 / 0.200 / 0.200 / 1.598 / 1.623 / 0.200 / 1.979	6.3932	4.951	178.73	1608.0
9021	W / Bi / U / Al / SiC / Fe / B	1.297 / 1.868 / 0.200 / 0.235 / 0.200 / 0.388 / 1.812	6.7280	18.485	147.76	10068.2
9022	W / Bi / U / Al / B / Fe / SiC	0.955 / 1.993 / 0.363 / 0.437 / 1.249 / 0.363 / 0.640	6.7438	6.519	147.98	7607.3
9023	Al / Bi / B / W / SiC / Fe / U	0.200 / 1.728 / 0.200 / 1.727 / 0.200 / 0.225 / 1.720	6.7073	8.693	143.85	13776.7
9024	W / Bi / U / Al / B / Fe / SiC	0.348 / 1.973 / 0.345 / 0.462 / 0.869 / 0.340 / 1.663	6.8905	1.079	146.66	3241.9
9025	W / Bi / U / Al / Fe / SiC / B	0.846 / 1.900 / 0.648 / 0.512 / 0.229 / 0.226 / 1.639	7.3786	17.720	129.27	6814.7
9026	W / Bi / B / Al / U / SiC / Fe	0.483 / 1.907 / 0.681 / 0.383 / 1.779 / 0.383 / 0.384	7.5245	4.042	126.04	4146.8
9027	B / SiC / Fe / W / Al / Bi / U	0.445 / 1.907 / 0.598 / 0.200 / 0.200 / 0.797 / 1.853	7.8751	4.952	138.24	1926.7
9028	W / Bi / SiC / Al / B / Fe / U	0.517 / 1.684 / 0.200 / 0.200 / 1.375 / 0.200 / 1.824	8.2276	4.776	108.25	4343.8
9029	W / SiC / Bi / B / Fe / U / Al	0.200 / 1.792 / 0.200 / 1.904 / 0.302 / 1.348 / 0.254	8.4727	6.184	117.89	1700.6
9030	Fe / Bi / SiC / Al / B / W / U	0.200 / 1.203 / 0.200 / 0.264 / 1.950 / 0.200 / 1.983	8.9462	4.883	95.15	2134.0

Table S10: Excluded ballistic records. Predictions are not valid recovered labels.

ID	Final E/E0	Max deviation	Outer Al cm	Predicted mm	Mass kg
733	0.69356	0.30647	0.289	13.208	210.6
832	0.8662	0.13388	0.267	16.444	179.33
1277	0.86747	0.13265	0.266	12.91	210.47
1567	0.93058	0.069796	0.261	3.9343	189.52

S6. Reproduction and boundaries of the revision

The revision scripts are `audit_coauthor_revision.py`, `revision_extra_diagnostics.py`, and `build_revision_assets.py`. They read the archived Phase 2–5 evidence and write only the revision directory. The main audit records SHA256 hashes of its input CSVs. The MCNP inventory records output hashes. Test bootstrap resampling uses seed 20261012. Lightweight baselines use training-only five-fold CV with seed 42. The MCNP-filter sensitivity clones the frozen Ridge pipeline and changes training eligibility only. It does not overwrite the serialized operational model or reselect its hyperparameters. The subsequently completed equal-budget optimizer study is documented in Section S8. Nested/repeated development splits, mesh convergence, and additional time extensions are not reported as completed analyses.

S7. Exact finite hyperparameter search spaces

The same five training folds (shuffle seed 42) are used for RandomizedSearchCV and development ranking. Twenty draws are requested, reduced to eight for the eight-value Ridge space. These are not nested-CV scores.

Ridge

- `alpha`: [0.0001, 0.001, 0.01, 0.1, 1.0, 10.0, 100.0, 1000.0].

RandomForest

- `n_estimators`: [300, 500, 700].
- `max_depth`: [None, 8, 12, 16, 24].
- `min_samples_leaf`: [1, 2, 4, 6].
- `max_features`: [0.5, 0.7, 0.9, 1.0].

ExtraTrees

- `n_estimators`: [300, 500, 700].
- `max_depth`: [None, 8, 12, 16, 24].
- `min_samples_leaf`: [1, 2, 4, 6].
- `max_features`: [0.5, 0.7, 0.9, 1.0].

GradientBoosting

- `n_estimators`: [200, 350, 500, 700].
- `learning_rate`: [0.015, 0.03, 0.05, 0.08].
- `max_depth`: [2, 3, 4].
- `min_samples_leaf`: [2, 4, 8, 12].
- `subsample`: [0.7, 0.85, 1.0].

HistGB

- `learning_rate`: [0.02, 0.04, 0.06, 0.1].
- `max_iter`: [200, 350, 500, 700].
- `max_leaf_nodes`: [15, 31, 63].
- `min_samples_leaf`: [10, 20, 30, 40].
- `l2_regularization`: [0.0, 0.01, 0.1, 1.0].

MLP

- `hidden_layer_sizes`: [(64, 32), (128, 64), (128, 64, 32), (256, 128, 64)].
- `alpha`: [1e-05, 0.0001, 0.001, 0.01].
- `learning_rate_init`: [0.0003, 0.001, 0.003].

XGBoost

- `n_estimators`: [300, 500, 700, 900].
- `max_depth`: [3, 4, 5, 6, 8].
- `learning_rate`: [0.02, 0.04, 0.06, 0.1].
- `subsample`: [0.7, 0.85, 1.0].
- `colsample_bytree`: [0.7, 0.85, 1.0].
- `reg_lambda`: [0.1, 1.0, 5.0, 10.0].
- `min_child_weight`: [1, 3, 5, 10].

S8. Equal-budget optimizer benchmark

The additional study comprises 100 runs and 6,030,000 candidate evaluations, using frozen dose and penetration pipelines. It is distinct from the original NSGA-II campaign used to select the thirty solver-re-evaluated configurations. Each condition uses ten paired seeds, 300 retained designs, and 60,300 evaluations per run. The lower limits are decision-threshold sensitivities only.

The main text reports aggregate results. Table S11 preserves all per-run outcomes. HV uses the fixed DoE anchors without clipping coordinates better than the DoE ideal. IGD and generalized spread use the pooled empirical nondominated reference for each limit/margin combination. At 60 mm, the margin reference pools three methods, whereas the point-only reference contains NSGA-II only. Consequently, identical NSGA-II fronts can have different IGD values in those two columns of the protocol. Such differences are reference-set effects, not different solutions.

Table S11: All optimizer runs. Q uses the frozen margin, P uses the point prediction. Times are elapsed seconds recorded by each run.

Method	Rule	Seed	HV	IGD	Spread	Time (s)	Limit (mm)
NSGA-II	Q	42	0.754124	0.036447	0.559936	1294.8	60
NSGA-II	Q	137	0.753940	0.035751	0.566454	1616.4	60
NSGA-II	Q	233	0.750328	0.033974	0.535882	1399.7	60
NSGA-II	Q	349	0.746614	0.037263	0.562335	1245.0	60
NSGA-II	Q	457	0.753255	0.035032	0.490775	1287.8	60
NSGA-II	Q	563	0.752529	0.034412	0.553291	1401.1	60
NSGA-II	Q	677	0.751071	0.034936	0.533257	1277.4	60
NSGA-II	Q	809	0.755110	0.035604	0.563552	892.9	60
NSGA-II	Q	929	0.753022	0.034289	0.574463	851.9	60
NSGA-II	Q	1049	0.755526	0.033577	0.521372	919.7	60
NSGA-II	P	42	0.754124	0.029998	0.559936	873.5	60
NSGA-II	P	137	0.753940	0.027246	0.566454	855.7	60
NSGA-II	P	233	0.750328	0.027037	0.535882	854.8	60
NSGA-II	P	349	0.746614	0.031445	0.562335	891.4	60
NSGA-II	P	457	0.753255	0.028654	0.490775	868.9	60
NSGA-II	P	563	0.752529	0.027146	0.553291	866.9	60
NSGA-II	P	677	0.751071	0.028734	0.533257	848.7	60
NSGA-II	P	809	0.755110	0.028503	0.563552	866.7	60
NSGA-II	P	929	0.753022	0.026997	0.574463	852.3	60
NSGA-II	P	1049	0.755526	0.027532	0.521372	7295.9	60
NSGA-II	Q	42	0.754487	0.026965	0.572185	861.6	15
NSGA-II	Q	137	0.753764	0.027827	0.516387	857.5	15
NSGA-II	Q	233	0.753038	0.030732	0.583810	855.2	15
NSGA-II	Q	349	0.754655	0.027015	0.553992	860.7	15
NSGA-II	Q	457	0.749742	0.028807	0.517877	870.1	15
NSGA-II	Q	563	0.754816	0.027930	0.552607	849.9	15
NSGA-II	Q	677	0.747401	0.028522	0.541477	848.7	15
NSGA-II	Q	809	0.756195	0.026396	0.563927	870.9	15
NSGA-II	Q	929	0.744968	0.030078	0.567020	860.5	15
NSGA-II	Q	1049	0.757689	0.026348	0.567346	855.0	15
NSGA-II	P	42	0.755297	0.029237	0.575146	861.2	15

Method	Rule	Seed	HV	IGD	Spread	Time (s)	Limit (mm)
NSGA-II	P	137	0.755748	0.027614	0.539429	874.5	15
NSGA-II	P	233	0.753730	0.028798	0.539091	856.8	15
NSGA-II	P	349	0.753522	0.027970	0.511500	854.6	15
NSGA-II	P	457	0.754674	0.027294	0.514396	856.6	15
NSGA-II	P	563	0.753708	0.027302	0.547777	869.9	15
NSGA-II	P	677	0.746563	0.029409	0.495057	858.9	15
NSGA-II	P	809	0.754487	0.027932	0.560630	860.7	15
NSGA-II	P	929	0.756881	0.028811	0.538881	859.3	15
NSGA-II	P	1049	0.743477	0.033467	0.551739	853.4	15
NSGA-II	Q	42	0.755460	0.029041	0.534081	862.4	20
NSGA-II	Q	137	0.751729	0.027947	0.484080	858.9	20
NSGA-II	Q	233	0.752217	0.028316	0.556126	855.2	20
NSGA-II	Q	349	0.755777	0.027190	0.493141	863.4	20
NSGA-II	Q	457	0.748049	0.029853	0.549583	858.8	20
NSGA-II	Q	563	0.749688	0.029589	0.572385	859.8	20
NSGA-II	Q	677	0.754219	0.027738	0.497498	855.1	20
NSGA-II	Q	809	0.746760	0.029539	0.541339	857.8	20
NSGA-II	Q	929	0.752422	0.028722	0.592711	865.7	20
NSGA-II	Q	1049	0.753014	0.030168	0.506040	1295.7	20
NSGA-II	P	42	0.751835	0.028540	0.545631	1033.8	20
NSGA-II	P	137	0.751691	0.028436	0.561064	924.1	20
NSGA-II	P	233	0.749067	0.029013	0.520167	871.5	20
NSGA-II	P	349	0.750255	0.028198	0.550876	876.6	20
NSGA-II	P	457	0.749316	0.027917	0.504636	881.9	20
NSGA-II	P	563	0.750883	0.028619	0.551583	980.9	20
NSGA-II	P	677	0.747130	0.031253	0.530193	1184.3	20
NSGA-II	P	809	0.758494	0.027310	0.520033	1215.6	20
NSGA-II	P	929	0.746392	0.028683	0.505894	1132.7	20
NSGA-II	P	1049	0.759523	0.027836	0.589500	1318.8	20
NSGA-II	Q	42	0.755754	0.028521	0.579656	1153.2	30
NSGA-II	Q	137	0.752010	0.027276	0.551448	1045.4	30
NSGA-II	Q	233	0.752657	0.026599	0.533920	1056.6	30
NSGA-II	Q	349	0.746886	0.030721	0.534174	1080.5	30
NSGA-II	Q	457	0.745212	0.029861	0.549814	961.9	30
NSGA-II	Q	563	0.752431	0.027938	0.463872	906.6	30
NSGA-II	Q	677	0.744596	0.030529	0.554947	906.0	30
NSGA-II	Q	809	0.744507	0.029753	0.543521	966.0	30
NSGA-II	Q	929	0.752280	0.027956	0.559346	886.5	30
NSGA-II	Q	1049	0.756932	0.027536	0.517209	895.0	30
NSGA-II	P	42	0.756656	0.027828	0.577484	881.4	30
NSGA-II	P	137	0.753547	0.026795	0.528283	893.5	30
NSGA-II	P	233	0.750328	0.028743	0.536244	887.4	30
NSGA-II	P	349	0.746614	0.030874	0.562112	893.4	30
NSGA-II	P	457	0.751527	0.027569	0.558931	886.8	30
NSGA-II	P	563	0.754788	0.028887	0.612648	887.2	30
NSGA-II	P	677	0.752680	0.027077	0.516221	888.5	30
NSGA-II	P	809	0.752512	0.028267	0.538344	897.8	30
NSGA-II	P	929	0.755528	0.028284	0.563204	886.3	30

Method	Rule	Seed	HV	IGD	Spread	Time (s)	Limit (mm)
NSGA-II	P	1049	0.755539	0.028790	0.573290	884.5	30
Random	Q	42	0.724130	0.041341	0.359173	510.5	60
Random	Q	137	0.722187	0.043888	0.393361	640.9	60
Random	Q	233	0.720216	0.043745	0.389232	600.6	60
Random	Q	349	0.722942	0.043948	0.402402	595.2	60
Random	Q	457	0.719920	0.045475	0.406610	594.6	60
Random	Q	563	0.722475	0.043412	0.413804	598.8	60
Random	Q	677	0.714935	0.047324	0.424122	517.4	60
Random	Q	809	0.719858	0.044130	0.386803	516.0	60
Random	Q	929	0.719796	0.043371	0.383962	527.6	60
Random	Q	1049	0.722042	0.042518	0.363235	518.7	60
NSGA-III	Q	42	0.760356	0.034044	0.490777	903.3	60
NSGA-III	Q	137	0.762360	0.033232	0.554325	911.7	60
NSGA-III	Q	233	0.766346	0.026225	0.484069	908.1	60
NSGA-III	Q	349	0.760388	0.033099	0.565579	919.4	60
NSGA-III	Q	457	0.759704	0.033400	0.560859	908.0	60
NSGA-III	Q	563	0.760782	0.036758	0.528910	896.6	60
NSGA-III	Q	677	0.766720	0.027139	0.451587	905.1	60
NSGA-III	Q	809	0.763811	0.032201	0.526008	915.7	60
NSGA-III	Q	929	0.755598	0.030910	0.527086	896.1	60
NSGA-III	Q	1049	0.763113	0.030454	0.507230	896.5	60

The spread of timings includes execution-condition variability and interruptions. The recorded sum is approximately 27.04 h, excluding time between stopped processes and work lost after the last checkpoint. It is not a controlled processor-time comparison, so these records do not establish that one optimizer is intrinsically faster.

Table S12 reports every prespecified primary contrast. Confidence intervals are paired-bootstrap intervals over ten seeds, not simultaneous intervals. The Holm-adjusted tests cover the family of six contrasts. Both method contrasts remain significant after correction. No margin contrast does. In

Table S12: Paired HV contrasts, A minus B. All rows use ten paired seeds.

Contrast	Mean difference	95% interval	Raw p	Holm p
II minus random, 60 Q	0.031702	[0.029448, 0.033562]	0.001953	0.011719
II minus III, 60 Q	-0.009366	[-0.011987, -0.006894]	0.001953	0.011719
Q minus P, 60 mm	0.000000	[0.000000, 0.000000]	1.000000	1.000000
Q minus P, 15 mm	-0.000133	[-0.003834, 0.003871]	1.000000	1.000000
Q minus P, 20 mm	0.000475	[-0.003186, 0.003702]	0.695313	1.000000
Q minus P, 30 mm	-0.002645	[-0.004878, -0.000509]	0.064453	0.257813

particular, the raw bootstrap interval excluding zero at 30 mm must not be described as multiplicity-adjusted evidence. Lack of significance does not prove equivalence.

The reproducibility entry point is `comparacoes_otimizacao_20261012.py`, launched through `EXECUTAR_COMPARACOES_20261012.ps1`. Its frozen plan records input/source hashes, seeds, library versions, and the exact evaluation budget. Each run retains its initial and final populations, final front, generation history, completion record, and checkpoint. The consolidated files include `comparison_per_run.csv`, `paired_HV_comparisons.csv`, and `margin_ablation_per_seed.csv`. The output named `near_boundary_candidates_NOT_HF_VALIDATED.csv` contains candidates for possible further analysis, not additional solver-confirmed cases.